

5

Built Environment

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- ***The Built Environment and Its Impact***
 - *Facilities management & Sustainable Facility management strategies*
 - *Building Control systems - Climate control systems, Security systems*
 - *Energy efficient built environments & Recycling*
 - *LEED Rating & other Indian rating systems*
 - *Intelligent/ Smart Buildings*
- ***Aesthetics of built environment***
 - *Role of Urban Arts Commissions*
 - *Heritage Conservation, Structural Repair and Rehabilitation, Retrofitting*
- ***Innovations and methodologies for ensuring Sustainability***

Besides giving a large number of multiple choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge. This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.

RATIONALE

This elementary unit on the built environment and its role to mitigate environmental impact, discusses the various practicalities, such as, Facility management, Building control systems, Building Rating, and Codes and Standards, to impart state-of-the-art knowledge to improve the civil engineers' technical know-how, and perhaps open up possibilities of pursuing future expertise in one of the several areas discussed.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U5-01: Knowledge on the various aspects of Built Environment and its impact

U5-02: Understanding of Aesthetics and its importance for cultural heritage

U5-03: Knowledge on innovations and methodologies for ensuring environmental Sustainability (Codes and Standards)

Unit-5 Outcomes	EXPECTED MAPPING WITH COURSE OUTCOMES						
	(1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)						
	CO-1	CO-2	CO-3	CO-4	CO-5	CO-6	CO-7
U5-01	3	3	2	2	3	3	2
U5-02	2	2	3	2	2	3	3
U5-03	3	3	3	2	3	3	3

Our engineered, built-environment sits nestled in the natural environment, which is presently undergoing several climatic changes leading to rising concerns and potential risks to human life. The built-environment is defined by the U.S. Environmental Protection Agency (EPA, 2021) as “human-made or modified surroundings in which people live, work, and recreate”, to which the National Academies (2017) adds, “surroundings that provide the setting for human activity, ranging in scale from buildings and parks or green space to neighbourhoods and cities that can often include their supporting infrastructure, such as water supply or energy networks”.

5.1 THE BUILT ENVIRONMENT AND ITS IMPACT

The major burden of societal and global impact is created and experienced at the intersection of “buildings, cities, and urban spaces, and the ways people use them” (WHO, 2016), i.e., intersection of ‘Spaces and Infrastructure’, comprising of design, construction, occupancy management, building operations, maintenance, furniture and equipment, etc.; and ‘People and Organisation’, consisting of health services, catering, event management, hospitality, security, safety, human resources, logistics, document management, and accounting. The concept of “**sustainable construction**” integrates environmental criteria into the entire lifecycle of a project, including building, maintenance, and eventual demolition, suggests various factors to be emphasized during the construction phase, beyond material choices, such as techniques to minimize dust, noise, and soil and water contamination, waste management measures throughout the construction, use, and demolition phases, and efforts to enhance material reuse and recycling.

Buildings alone contribute to approximately 40% of energy and process-related CO₂ emissions, 50% of all extracted materials, 33% of water consumption, and 35% of generated waste, and additionally, contribute towards resource depletion, pollution of air, water, and land, as well as loss of biodiversity. Of which, the use/operational phase leads to the highest environmental impact across the entire life of the building and is known to be caused by the users and the physical characteristics of the building, i.e., energy and water use and waste generation. Therefore, it is important to employ *systems thinking and the lifecycle view* to address impending issues of environmental impact mitigation through incorporating sustainability strategies, beyond design and construction, into facility management.

5.1.1 Facilities Management

Facilities management is a vital aspect of managing construction businesses, as it involves strategic planning and thoughtful consideration of daily operations. British architect and Editor of Architectural Association Journal (1965-67) and Facilities (1982-90), Francis Duffy explored the relationship between organisational structure and office layouts in his doctoral work during the 1960s, and later, pioneered ‘Space Planning and Facility management’ in Europe. At the

time, the term was propounded by Ross Perot, founder of Electronic Data Systems, USA, when the introduction of computers and IT infrastructure demanded design changes in the 1970s.

Facility management entails the effective management of physical workspaces - buildings, infrastructure, and other physical assets, to support the delivery of services and achieve organizational objectives. Civil engineers are ideal for this role due to their technical skills and good communication, and develop leadership in the upcoming area of facility management. Facility Management integrates the principles of business administration, architecture, and the behavioural and engineering sciences; and involves a wide range of activities, including maintenance and operations, space planning, asset management, health and safety, energy management, and sustainability.

The **key activities** involved are:

- Asset Management
- Master Planning
- Space Management
- Estates Strategies
- Maintenance, cleaning, testing, and inspection
- Refurbishment, retrofitting, and renovation
- Acquisitions and sales
- Procurement and project management
- MEP (Mechanical, Electrical and Plumbing) and Technical services
- Contract Management
- Sustainability
- Budget Management
- Brand Management
- Quality Assessment
- Regulatory Compliance
- Asset exploitation and income generation
- Ensuring the continuity of the business
- Delivery of new technologies
- Bringing change and efficiency in work practices
- Safety and security
- Traffic, transport, and parking

Role of Facility manager responsibility towards sustainability

In the field of facilities management, the focus has shifted to the impacts of climate change, encompassing not only the environmental effects of buildings but also their resilience and capacity to address associated issues. The primary challenge lies in the increasingly severe weather conditions and unpredictable shifts in weather patterns, such as tides, cold spells, floods, storms and droughts, and prolonged seasons, making it necessary for facilities managers to transition from preventive measures to predictive strategies, like enhanced service level

requirements and disaster recovery plans for ensuring business continuity. Facility managers have **three significant responsibilities** in addressing climate change:

- **Conduct assessments to evaluate the vulnerability of current facilities** and their ability to adapt to climate change. This involves creating plans and procedures to address climate-related challenges and ensure effective responses.
- **Assist in the development of new structures and buildings** that align with the changing environmental landscape. This includes reviewing designs and incorporating sustainable building materials to promote environmentally friendly practices.
- **Develop a forward-looking strategy to manage the evolving dynamics of the built environment** due to shifts in climate change regulations and compliance requirements. This involves anticipating future changes and implementing measures to ensure ongoing environmental sustainability.



Fig. 5.1 : 11 Core competencies of Facility Management (source : www.ifma.org)

Sustainability Facility management strategies

Sustainable facility management (SFM) is “a unique process that offers a facility manager the authority to make structural, architectural, and operational changes to reduce the negative impact of buildings on their occupants and the environment” (Fennimore, 2014). Implementing sustainable practices in facility management to mitigate climate change risks requires a comprehensive approach that considers various aspects of operations. Adopting a lifecycle approach in decision-making processes, taking into account the environmental impacts of products and systems from extraction to disposal, and considering factors such as, durability, recyclability, and end-of-life management are essential towards enabling sustainable facility management.

The following are some **practical strategies** for implementing sustainable practices in facility management:

1. **Energy Efficiency:** Implement energy-saving measures, such as, installing energy-efficient lighting, utilizing programmable thermostats, optimizing HVAC systems, and conducting energy audits to identify areas for improvement.
2. **Renewable Energy Integration:** Explore opportunities to incorporate renewable energy sources, such as solar panels or wind turbines to reduce reliance on fossil fuels and decrease carbon emissions.
3. **Waste Management:** Develop a waste management plan that includes recycling programs, waste reduction initiatives, proper disposal methods, and encourage all stakeholders to reduce, reuse, and recycle materials.
4. **Water Conservation:** Implement water-saving measures, such as, installing low-flow fixtures, conducting regular maintenance to prevent leaks, and promoting water-efficient practices among all stakeholders.
5. **Indoor Air Quality (IAQ):** Implement measures to improve indoor air quality, such as, regular maintenance of HVAC systems, proper ventilation, and use of low VOC (volatile organic compound) materials in building interiors.
6. **Green Building Certification:** Consider pursuing green building certifications, like LEED (Leadership in Energy and Environmental Design), to guide sustainable building practices and ensure compliance with established standards.
7. **Performance Monitoring:** Establish performance metrics, and regularly monitor and track energy consumption, water usage, waste generation, and other sustainability indicators, and in turn, use this data to identify areas for improvement and set targets for continuous enhancement.
8. **Sustainable Procurement:** Source environmentally friendly products and materials that are made from recycled content, are energy-efficient, and have a reduced environmental impact. Consider the lifecycle of products and choose those with minimal environmental footprints.
9. **Stakeholder Engagement:** Engage employees, tenants, and other stakeholders in sustainability initiatives by promoting awareness, providing training on sustainable practices, and encouraging their active participation.

5.1.2 Building Control system

A building control system is a centralized system that monitors and controls various building systems and equipment to ensure efficient operation, occupant comfort, and safety. These systems are commonly referred as; Energy Management Systems (EMS), which is responsible for managing environmental functions within a building; Building Automation Systems (BAS), which focuses on controlling technical automation processes; and Building Management Systems (BMS), which encompasses a broader range of functions, including providing status reports on environmental

conditions, monitoring elevator operations, and tracking the location of individuals for security purposes.

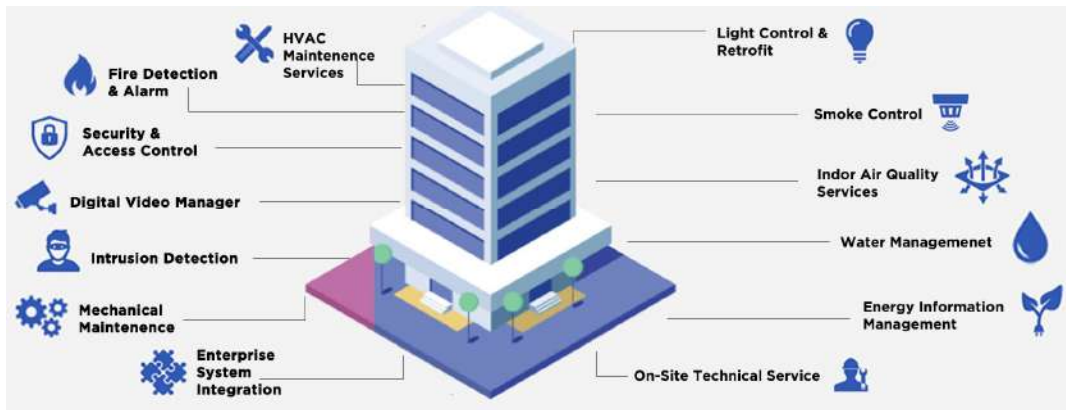


Fig. 5.2: Elements of a Building Control System (source : www.truiconline.com)

The major functions are, as follows;

- **Data Analysis and Reporting:** Building control systems provide data analysis and reporting capabilities, enabling facility managers to assess building performance, identify trends, and make informed decisions for improving energy efficiency and operational effectiveness.
- **Monitoring and Control:** Monitoring systems collect data on various parameters such as temperature, humidity, occupancy, and energy consumption to monitor the building's performance and ensure it operates within predefined limits. Based on the collected data, the system can automatically adjust and control the building systems to maintain optimal conditions. For example, it can regulate HVAC systems to provide comfortable temperatures or adjust lighting levels based on occupancy.
- **Management and Decision-aid:** Building control systems help optimize energy usage by tracking energy consumption, identifying inefficiencies, and implementing strategies for energy conservation, such as scheduling equipment operation and implementing demand response programs. The system also generates alerts and notifications in case of equipment malfunctions, abnormal conditions, or security breaches, allowing facility managers to take prompt action.

EMS, BAS, and BMS consists of a network of sensors, controllers, and software that communicate and coordinate the functioning of different building systems. EMS collects data from sensors throughout the building and uses the information to make adjustments and optimize the performance of these systems. While BMS/BAS typically integrates and controls systems such as heating, ventilation, and air conditioning (HVAC), lighting, access control, fire alarms, security systems, and energy management. The key difference is, while an EMS is primarily a monitoring tool at a micro-level, gathering detailed device-level telemetry data from the deployed sensors and analysing it, the BMS/BAS, often used synonymously, works at a macro-

level by providing high-level, real-time control, such as, regulate the temperature, humidity or pressure, or prevent smoke from spreading throughout a building.

Building Climate Control System

A building and its indoor spaces are constantly influenced by both internal and external factors that can disrupt the indoor climate, such as heat, air emissions, and changes in humidity levels. Climate control in buildings refers to the adjustment of indoor climate conditions according to the preferences of the occupants, while ensuring that external weather conditions, such as extreme heat or harsh environmental factors, do not impact the indoor environment. **Indoor Environment Quality (IEQ)** refers to “*the quality of a building’s environment in relation to the health and wellbeing of those who occupy space within it*”, as per the National Institute of Occupational Safety and Health (NIOSH). IEQ is of great importance, as the four primary parameters - thermal, visual, and acoustic comfort, indoor air quality (IAQ), have *direct impact on various aspects of human well-being, such as, occupant comfort, productivity, mental health, satisfaction, as well as study and work performance* (Ali and Acharya, 2023) and hence, there is a need for climate control considering the drastic change in temperatures, high levels of noise and particulate pollutants in the surrounding and excess use of artificial lighting that may cause visual discomfort.

Architectural design, engineering and construction is of utmost importance to ensure climate control and comfortable IEQ as several factors such as, temperature, humidity, rainfall, direction of wind and sunlight, etc. plays a crucial role in material selection, orientation and structural design. Air temperature affects the exterior of the building while the soil temperature affects the foundation. Humidity is a very important consideration for the structural integrity of the materials, and anti-corrosive and water-resistant materials are preferred in areas with higher levels of humidity. In areas with heavy rainfall, sloping roofs are preferred as opposed to flat roofs, so that the integrity of the structure is not compromised. Sunlight is also an important consideration, as the wall to window ratio increases or decreases according to need of the climate. Wind load is also considered while designing, since it has the capability to damage the building structurally. **Active and Passive design strategies** are employed to improve IEQ and liveability of a building. ‘Active’ implies the use of mechanical and electrical systems, such as, air-conditioning, heat pumps, radiant heating, heat recovery ventilators, and electric lighting which uses purchased energy, while ‘Passive’ refers to harnessing available ambient energy through techniques such as, maximizing natural light (daylighting), utilizing natural airflow for ventilation, and garnering solar energy. Some **passive design strategies** are;

- **Seeking inspiration from Vernacular architecture.** It is the ancient traditional architectural design and construction method of a civilisation, and usually it is incorporated with unique elements and techniques that controls the natural climate of that region and suitable of the life of the indigenous people. Introduction of verandahs and courtyards are one such design element borrowed from vernacular architecture that has the ability to reduce heat and allow cross-ventilation.

- **Incorporating Bioclimatic design**, which aims to maximize the utilization of local climate data and leverages local environmental resources like sunlight, wind, air, vegetation, soil, and sky to minimize energy consumption. Proper thermal insulation of the external envelope of the building and air-tight structures; use of solar energy for heating in winter time, achieved by a proper passive solar heating system; removal of the heat from the building by passive cooling systems and natural ventilation; Protection of the building from overheating by proper shading solutions and vegetation in summer time; and the use of solar energy for day-lighting all year round, are some of the recommendations by EU. It may be argued that this is also seen in vernacular architecture.
- **Employing shading devices**, such as, fin walls, vertical or horizontal shading overhang devices, and extended roof overhangs, which are integrated into the building structure, and external shading devices, such as, eaves, awnings, screens and shutters, louvres, verandahs, pergolas, and shading with foliage, trees, and shrubs.
- **Introducing rigid air barrier, thermal insulation and sound attenuating interior lining** to brick or concrete masonry or to high-density sheet cladding for exterior walls and building envelope, using concrete or tiled roofing, and affixing sound-attenuating external windows and doors, to reduce external noise and overall, acoustic discomfort.

‘Active’ control systems leverage technologies, such as, ICT, cloud computing, wireless sensors, and tools, such as, BMS and BAS to help improve climate control efficiency.

Building Security System

The **building security system** is specifically developed to oversee and regulate mechanical and electrical installations, fire protection and escape mechanisms, as well as address issues related to burglary, assault, and emergency communication.



Fig. 5.3 : Features of a Building Security System (source:www.truviconline.com)

Unit 5 – Built Environment

In high-rise buildings and large complexes which are densely occupied, ensuring effective fire safety measures is of utmost importance. To ensure **fire safety** as it is necessary to monitor and manage the following:

- Detection and suppression of fires,
- Ensuring the movement and safety of individuals,
- Controlling smoke through pressurization and barriers,
- Establishing safe places of refuge, and
- Facilitating emergency arrangements and communication.

An effective security system has the following components:

Video surveillance systems comprises of the placement of multiple cameras in strategic locations both inside and outside a building, to serve as a deterrent to potential criminals, and to provide valuable evidence for police investigations and insurance claims. Furthermore, modern video surveillance systems can be integrated with other security technologies. For instance, you can have a surveillance camera positioned to monitor a keypad entry system at the front door, which can enable detection of any tampering or misuse of the entry device.

Access control encompasses the procedures and mechanisms that regulate entry and restrict unauthorized access to buildings, rooms, or specific areas, and is a fundamental aspect of maintaining security in any location. To ensure optimal security, it is essential to have a cohesive access control system in place, as it is inconvenient for occupants to use different Access control technologies, such as, separate key fobs, PIN codes, or key cards, to access different areas within the building. Therefore, it is recommended to synchronize the access control system throughout the building, preferably by using a unified system provided by the same security system company.

Alarm systems are installed at doors and windows, including emergency-only exits, and are designed to emit a loud noise when triggered, alerting building staff to the specific location. If empowered with smart technology it can notify occupants about the occurrence and precise location of the triggered alarm, enhancing response effectiveness.

Intercom systems provide a means for visitors to request entry into the building, while allowing tenants to remotely grant access, and facilitates verification of the identity of individuals seeking entry.

Computer systems are essential to cohesively tie up the surveillance, access control, alarm and intercom systems in the backend and enable the user to control, communicate and interact remotely with assets and people. Currently, smartphones are integrated to allow the same, on the move.

5.1.3 Energy Efficient Built Environment

An **energy efficient building** “offers an appropriate environment for habitation with minimal energy consumption and wastage of energy, thereby maximizing energy conservation” and “balances all aspects of energy use in a building by providing an optimized mix of passive solar–design strategies, energy-efficient equipment, and renewable sources of energy” (Sustainable Fuel Technologies Handbook, 2021).

Life Cycle Assessment for Building and its materials

Life Cycle Assessment (LCA) based on ISO14040 (2006) is being increasingly adopted in architectural design and construction, as it CA enables decision-making processes, such as, evaluating and selecting environmentally-friendly products and optimizing construction processes. In addition to assessing environmental impacts, LCA also facilitates estimating life cycle costs (LCC) and conducting life cycle energy analysis (LCEA). The incorporation of LCA into the building sector supports sustainable decision-making and promotes the evaluation of environmental, economic, and energy-related aspects throughout the life cycle of buildings. It is noteworthy that energy efficiency has been a primary focus in the environmental design of buildings, with the **use/operational phase** playing a dominant role in the LCA of buildings, due to the high energy demand associated with building operation. Operating energy, which includes energy for HVAC, domestic hot water, lighting, appliances, and building maintenance, is easily quantifiable and can account for up to 85% of total energy consumption and 70-90% of the environmental impact. However, many of these factors are beyond the control of designers, particularly during the conceptual stage, as they are spread over the decades-long lifespan of a building.

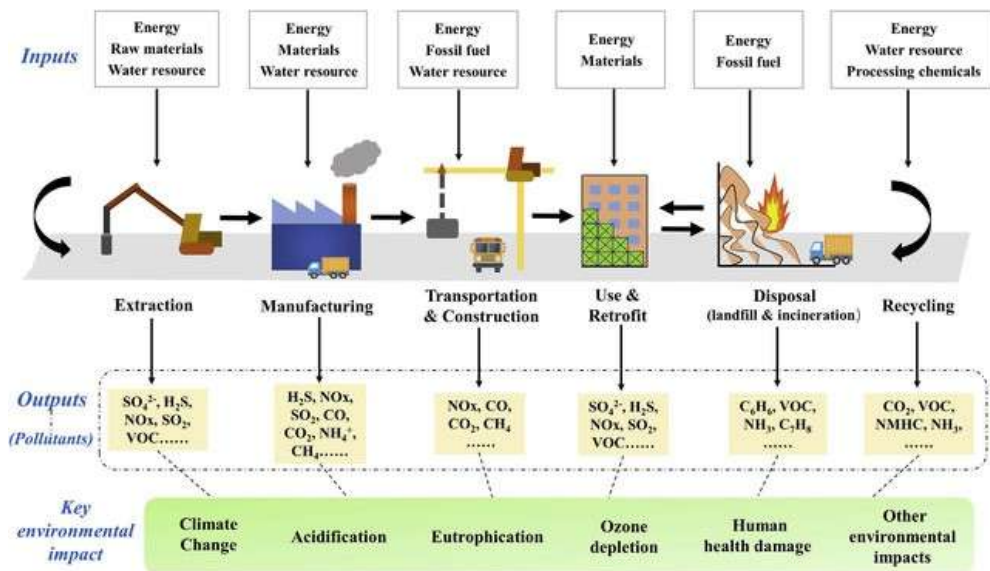


Fig. 5.4 : Key Environmental Impacts during Life Cycle of Building Materials (Huang, et al., 2020)

Therefore, the selection of appropriate materials, which is often emphasized as a directive for energy-efficient design, becomes a critical decision factor in the design process. It is recommended to consider material selection as early as possible, as it not only impacts the operational energy performance of the building but also influences the total embodied energy and potential environmental impact.

Embodied Energy and Embodied Carbon

From a life cycle perspective, the environmental impact and energy consumption of buildings are closely linked to the choice of materials. However, there is often a lack of alignment between the service life of materials (which determines the intervals for material renewal) and the service life of buildings (the duration of the operational phase). This can result in a *trade-off between material selection and building energy demand* and so, requires a holistic approach (Heeren et al., 2015) that considers the effects and trade-offs. For example, in a residential building, since the ‘wood variant’ is susceptible to solar and internal gains, and the ‘concrete variant’ isn’t, the energy spent on demand for space cooling of wood is higher than concrete in spite of wood having lower environmental impact than concrete. However, for cool climates, a wooden structure and this very same property will be beneficial and energy-saving.



Every building material possesses an ‘**Embodied Energy**’ or **Embodied Carbon**, which refers to the sum impact of all greenhouse gas emissions attributed to that material during its life cycle from extraction, till disposal. The after-use life cycle phase plays a significant role in determining the overall embodied energy of a building. LCA for the "end of life" (EOL) evaluates the environmental impact, including the energy required for demolition, at the end of a building's service life, as this is vital for decision-making since it has the potential to substantially reduce the environmental impact of the building through effective after-use strategies, such as, disassembly, remanufacturing, biodegrade-ability and the 3R principles of reduce, reuse, and recycle .

Lupíšek, et al. (2015) innumerate the **Design strategies** for reduction of embodied energy and embodied carbon (Subtask 4 of Annex 57) in three steps:

1. ***Reduction of amount of needed materials throughout entire life cycle***
 - 1.1. Optimization of layout plan
 - 1.2. Optimization of structural system
 - 1.3. Low-maintenance design
 - 1.4. Flexible and adaptable design
 - 1.5. Components’ service life optimization
2. ***Substitution of traditional materials for alternatives with lower environmental impacts***
 - 2.1. Reuse of building parts and elements
 - 2.2. Utilization of recycled materials
 - 2.3. Substitution for bio-based and raw materials
 - 2.4. Use of innovative materials with lower environmental impacts

- 2.5. Design for deconstruction
- 2.6. Use of recyclable materials

3. *Reduction of construction stage impact*

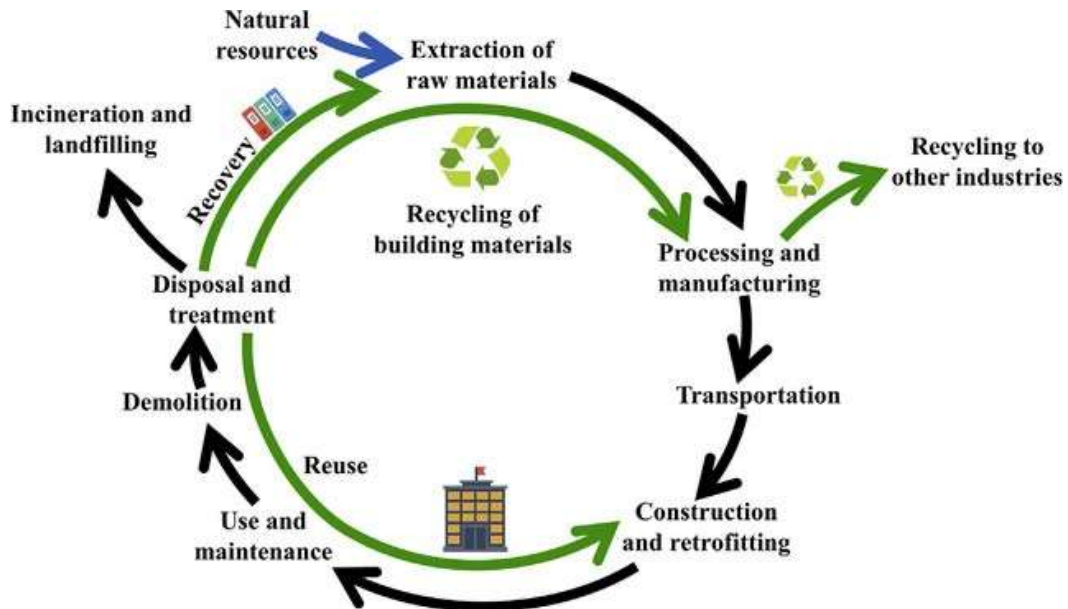


Fig. 5.5: Recycling of Building Materials (Huang, et al., 2020)

Building Materials and Recycling

A large number of building materials are **reusable and recyclable** (Kralj & Marič, 2008) , such as, Wood and untreated timber, and Earthen Materials (*reusable/recyclable/biodegradable*); Metals, mainly Steel, Aluminium, Iron, Copper, Masonry and Bricks, and Glass and ceramic (*reusable/recyclable*); Asphalt and Concrete (*may be crushed and recycled*); and Gypsum/Drywall (*recyclable, sometimes biodegradable*). Some innovative, modern recycled building materials are ; **Plant-Based Polyurethane Foam Boards**, possessing good heat transfer resistance, quality sound insulation, and mould and insect resistance; **Hempcrete**, a sustainable, carbon negative alternative to concrete bound with hemp fibres; **Ecobrick**, a plastic bottle filled with small plastic waste; **Rammed Earth**, made of compacted, excavated earth from the construction site, mixed with concrete; **Chip[s]** board, is a fibre-reinforced bioplastic made from potato waste.

Recycling materials can contribute greatly towards **reducing impact** and can be achieved by;

- **Identifying materials in existing buildings** that can be salvaged and reused in new construction projects may include, structural elements, fixtures, flooring, doors, and windows.
- **Establishing on-site recycling programs** to process and reuse construction and demolition waste, and setting up in dedicated areas for sorting and separating different materials such as concrete, metal, wood, and plastics.
- **Implementing comprehensive waste management plans** for construction sites, with clear guidelines for sorting, separating, and recycling different types of waste generated during the construction process.
- **Participating in material exchange networks** or online platforms where builders, contractors, and suppliers can connect to exchange surplus materials.
- **Collaborating with local recycling centers**, waste management facilities, and other stakeholders to establish efficient recycling systems for construction materials.

Incorporating design principles of ‘**Design for Disassembly**’ make it easier to disassemble and separate materials during the end-of-life phase, by using modular construction techniques and joinery systems, for easy dismantling without damaging the materials, is a potential approach to carefully deconstruct buildings to preserve reusable materials.

5.1.4 LEED Rating

LEED, which stand for Leadership in Energy and Environmental Design, is a green building rating system developed by the US Green Building Council (USGBC). In India, Confederation of Indian Industry (CII) formed the Indian Green Building Council (IGBC) in year 2001 and has licensed the LEED Green Building Standard from the USGBC.

Policymakers, planners, and builders can adopt LEED and sustainable building practices as effective strategies to progress towards the UN SDGs. LEED categories contribute to the achievement of the SDGs in the following ways - conserving water, improving energy efficiency, reducing carbon emissions (GHGs), and minimizing harmful air pollutants to enhance environmental sustainability. LEED also supports education, job creation, improved health and well-being, strengthened community resilience, and more. It takes a holistic approach by considering all essential aspects that contribute to the overall quality of a building, rather than solely focusing on individual elements like energy, water, or health, and embraces a comprehensive perspective, recognizing the interplay and synergy among various factors to achieve optimal building performance.

The **primary objective** of LEED is to facilitate the design of good buildings that exude the following characteristics, as outlined by USGBC;

- *Reduce contribution to global climate change.*

- *Enhance individual human health.*
- *Protect and restore water resources.*
- *Protect and enhance biodiversity and ecosystem services.*
- *Promote sustainable and regenerative material cycles.*
- *Enhance community quality of life.*

Certification types and Credit scoring

There are **various kinds of LEED certifications** depending on the types and the phase of the building of projects, as follows;

- **LEED Building Design and Construction (BD+C)** For new construction or major renovations, and also includes applications for Schools, Retail, Hospitality, Data Centers, Warehouses & Distribution Centers, and Healthcare.
- **LEED Interior Design and Construction (ID+C)** for complete interior fit-out projects, Includes Commercial Interiors, and also includes applications for Retail and Hospitality.
- **LEED Building Operations and Maintenance (O+M)** for existing buildings that are undergoing improvement work or little to no construction. Includes Existing Buildings, and also includes applications for Schools, Retail, Hospitality, Data Centers, and Warehouses & Distribution Centers.
- **LEED Neighbourhood Development (ND)** for new land development projects or redevelopment projects containing residential uses, non-residential uses, or a mix. Projects can be at any stage of the development process, from conceptual planning to construction. Includes Plan and Built Project.
- **LEED Homes** for single family homes, low-rise multi-family (one to three stories) or mid-rise multi-family (four or more). Includes Homes, Multifamily Low-rise, Multifamily Midrise.
- **LEED Cities** for entire cities and sub-sections of a city. LEED for Cities projects can measure and manage their city's water consumption, energy use, waste, transportation and human experience.

To obtain **LEED certification**, a project accumulates points by adhering to prerequisites and fulfilling criteria related to carbon emissions, energy efficiency, water conservation, waste management, transportation, materials usage, and indoor environmental quality. The certification process involves a thorough examination and assessment by GBCI (Green Business Certification Inc.), which assigns points based on the project's adherence to LEED standards. The total number of points determines the level of LEED certification: *Certified* (40-49 points), *Silver* (50-59 points), *Gold* (60-79 points), or *Platinum* (80+ points). Notably, climate change accounts for 35% relate to climate change, 20% directly impact human health, 15% impact water resources, 10% affect biodiversity, 10% relate to the green economy, and 5% impact community and natural resources. In LEED v4.1, operational and embodied carbon have the most LEED credits associated to them.



LEED Credits and categories

The **major categories and their respective credits** are;

- ***Energy and Atmosphere (33 credits)***
 - Fundamental Commissioning and Verification [*prerequisite*]
 - Minimum Energy Performance [*prerequisite*]
 - Building-Level Energy Metering [*prerequisite*]
 - Fundamental Refrigerant Management [*prerequisite*]
 - Optimize Energy Performance [18]
 - Enhanced Commissioning [6]
 - Advanced Energy Metering [1]
 - Renewable Energy [5]
 - Enhanced Refrigerant Management [1]
 - Grid Harmonization [2]
- ***Location and Transportation (16 credits)***
 - Sensitive Land Protection [1]
 - High Priority Site and Equitable Development [2]
 - Surrounding Density and Diverse Uses [5]
 - Access to Quality Transit [5]
 - Bicycle Facilities [1]
 - Reduced Parking Footprint [1]
 - Electric Vehicles [1]
- ***Materials and Resources (13 credits)***
 - Storage and Collection of Recyclables [*prerequisite*]
 - Building Life-Cycle Impact Reduction [5]
 - Environmental Product Declarations [2]
 - Sourcing of Raw Materials [2]
 - Material Ingredients [2]
 - Construction and Demolition Waste Management [2]
- ***Sustainable Sites (10 credits)***
 - Construction activity pollution prevention [1]
 - Site Assessment [1]
 - Protect or Restore Habitat [2]
 - Open Spaces [1]
 - Rainwater Management [3]
 - Heat Island Reduction [2]
 - Light Pollution Reduction [1]
- ***Water efficiency (10 credits)***
 - Outdoor Water Use Reduction [*prerequisite*]

- Indoor Water Use Reduction [*prerequisite*]
- Building-Level Water Metering [*prerequisite*]
- Outdoor Water Use Reduction [2]
- Indoor Water Use Reduction [6]
- Optimize Process Water Use [2]
- Water Metering [1]
- **Innovation (6 credits)**
 - Innovation in design [5]
 - LEED Accredited Professional [1]
- **Regional Priority Credits (4 credits)**
- **Integrative Process (1 credit)**

In the US Green Building (USGBC) annual list for **2021, India** achieved the remarkable position of **third in the world for LEED** Green buildings, with a total of 146 certified buildings and spaces, covering an impressive 2.8 million gross area square meters.

Other Green Building Certifications in India

Apart from LEED, the other 2 main green building certification systems in India, are; GRIHA (Green Rating for Integrated Habitat Assessment), India's own rating system jointly developed by TERI and the Ministry of New and Renewable Energy, Government of India, and BEE (Bureau of Energy Efficiency) developed its own rating system for the buildings and the Energy Performance Index (EPI).

GRIHA introduces a comprehensive approach of assessing 11 Parameters with 30 criteria accumulating to 100+5 maximum points (refer Fig. 6), across three stages:

1. **Pre-construction stage:** This stage addresses both on-site and off-site factors, such as the proximity to public transportation, soil type, land characteristics, location, existing flora and fauna, and natural landscape features before construction begins.
2. **Building planning and construction stages:** This stage focuses on resource conservation, reducing resource demand, optimizing resource utilization efficiency, promoting resource recovery and reuse, and ensuring occupant health and well-being.
3. **Building operation and maintenance stage:** This stage encompasses the operation and maintenance, monitoring and recording energy consumption, ensuring occupant health and well-being, and addressing factors that impact the local and global environment.

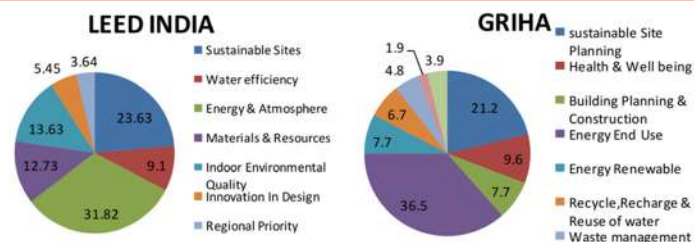


Fig. 5.6 : Performance criteria comparison of LEED and GRIHA Rating (Phadtare and Sande, 2015)

Unit 5 – Built Environment

Ji et al., 2017, conducted a comparative analysis of GRIHA v2015 and LEED v4 for BD + C: New Construction and Major Renovation with respect to the three pillars of Sustainability and reported;

GRIHA v2015	LEED v4 BD+C
Strong focus on Environmental and Economic aspects, less Social	Has more balanced focus on all the three pillars
Has a dedicated section for assessment of socio-economic aspects of the laborers and service professionals, creating awareness on environment and design for accessibility, missing from LEED.	Has a section dedicated to <i>Regional Priority: Specific Credit</i> , and has social aspects like, open space access, bicycle facilities, diversity in use and quality views included in its assessment. Can be an addition to GRIHA.

GRIHA v.2019			
Section	Criterion No.	Criterion Name	Maximum Points
1. Sustainable Site Planning	1	Green Infrastructure	5
	2	Low Impact Design	5
	3	Design to Mitigate UHIE	2
2. Construction Management	4	Air and Soil Pollution Control	1
	5	Top Soil Preservation	1
	6	Construction Management Practices	2
3. Energy Efficiency	7	Energy Optimization	12
	8	Renewable Energy Utilization	5
	9	Low ODP and GWP Materials	1
4. Occupant Comfort	10	Visual Comfort	4
	11	Thermal and Acoustic Comfort	2
	12	Maintaining Good IAQ	6
5. Water Management	13	Water Demand Reduction	3
	14	Wastewater Treatment	3
	15	Rainwater Management	5
	16	Water Quality and Self-Sufficiency	5
6. Solid Waste Management	17	Waste Management-Post Occupancy	4
	18	Organic Waste Treatment On-Site	2
7. Sustainable Building Materials	19	Utilization of Alternative Materials in Building	5
	20	Reduction in GWP through Life Cycle Assessment	5
	21	Alternative Materials for External Site Development	2
8. Life Cycle Costing	22	Life Cycle Cost Analysis	5
9. Socio-Economic Strategies	23	Safety and Sanitation for Construction Workers	1
	24	Universal Accessibility	2
	25	Dedicated Facilities for Service Staff	2
	26	Positive Social Impact	3
10. Performance Metering and Monitoring	27	Commissioning for Final Rating	7
	28	Smart Metering and Monitoring	0
	29	Operation and Maintenance Protocol	0
Total Points			100
11. Innovation	30	Innovation	5
Grand Total Points			100 + 5

Fig. 5.7 : GRIHA v2019 Criteria and Points (source : www.grihaindigo.org)

5.1.5 Intelligent to Smart Buildings

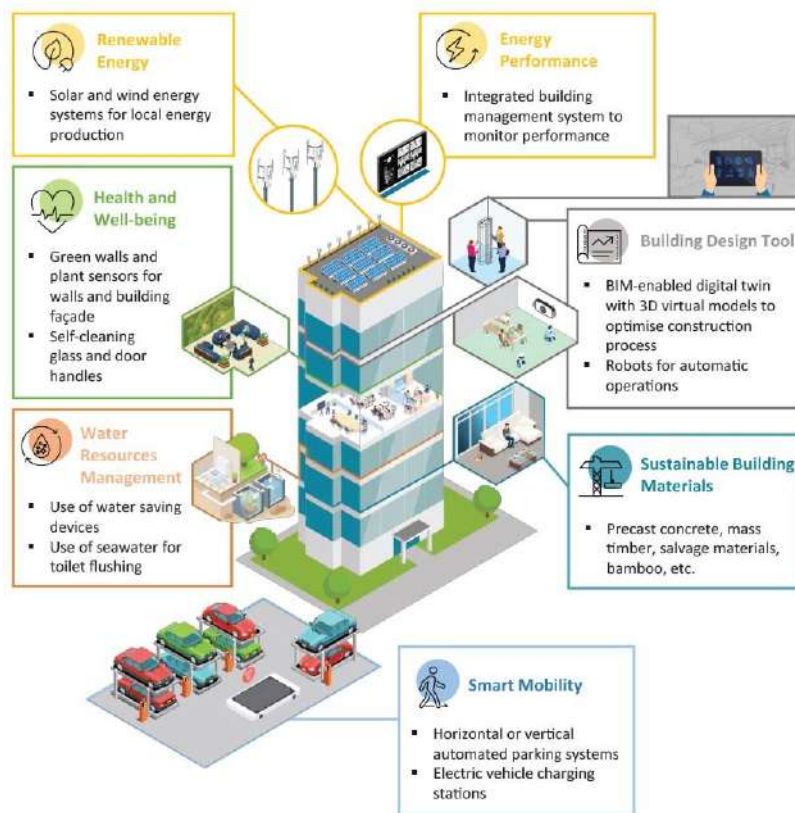


Fig. 5.8 : Smart Building technologies and features

The term, ‘Intelligent building’, is often used commonly to imply and ‘Smart building’, but their definitions have been evolving with time, since the 1980s when the term was first used to mean “*a building which totally controls its own environment*” (Stubbings, 1988). This equated the building to its BMS and other smart systems, with the intent to reduce human interaction. However, the two terms are complementary, as it may be argued that **Smart Buildings are Intelligent Buildings but with additional, integrated aspects** of ‘*adaptable control, enterprise and materials and construction*’, and addresses both ‘intelligence and sustainability issues’ to achieve the optimal energy consumption and overall comfort.

Smart Buildings are viewed as a ‘subset of smart environments’, where smart environments are “*able to acquire and apply knowledge about the environment and its inhabitants in order to improve their experience in that environment*”. In a **smart environment**, such as ‘Smart City’, various elements, i.e., Smart Homes, Smart Buildings, Smart Grids, Smart Meters, Smart Transportation, etc. collaborate synergistically. Smart Buildings prioritize the occupants and

encourage their active participation by incorporating feedback mechanisms, for both receiving and providing information about building usage. These buildings leverage integrated enterprise and intelligent systems to enable inherent control, by empowering occupants to make decisions regarding their comfort, while still maintaining regulated control. Smart buildings promote a personalized and interactive building experience.

Literature states that Smart Buildings have the **five fundamental** features;

- **Automation**: the ability to accommodate automatic devices or perform automatic functions.
- **Multi-functionality**: the ability to allow the performance of more than one function in a building.
- **Adaptability**: the ability to learn, predict and satisfy the needs of users and the stress from the external environment.
- **Interactivity**: the ability to allow the interaction among users.
- **Efficiency**: the ability to provide energy efficiency and save time and costs.

Dakheel, et al. (2020) analysed the above and extended it with four ‘Smartness basic features’, categorised with respect to four key functions - **Climate response**, **Grid response**, **User response** and **Monitoring and Supervision** - to include;

- **Nearly zero energy buildings target** (*Climate response, Grid response*)
- **Flexibility** (*Climate response, Grid response, User response*)
- **Real-time monitoring** (*Monitoring and Supervision*)
- **Real-time Interaction** (*User response*)

However, every building has a capacity to adapt their operation to the needs of the grid and the occupants, and this is measured by **Smart-readiness indicator** (SRI), developed by EPBD. The SRI in buildings (Verbeke et al., 2017) are particularly important for India as we have stark socio-economic diversity, and introducing the right amount of ‘*smartness*’ holds immense potential for transformative advancements in efficiency, sustainability, occupant comfort, and overall building performance.

5.2 AESTHETICS OF BUILT ENVIRONMENT

Aesthetics, a field within philosophy, is the study of the nature, expression and principles of beauty and artistic taste. The term originates from the Greek word "*aisthetikos*", which refers to sensory perception and understanding, or knowledge acquired through the senses, and hence, varies across people, places and time. There is no one style for aesthetics, but a constant evolution of elements as per the socio-cultural need of that time, or as discussed in earlier Units, as a response to an existing style. It is a deeper, more profound expression of human creativity, meant to be evocative, thought provoking or awe-striking. The **Vitruvian triad** – *firmitas* (durability), *utilitas* (usefulness), and *venustas* (beauty), deemed essential for a good building, captured the significance of aesthetics as

visual appeal encompasses various design elements and principles, such as, its form, size, texture, colour, balance, unity, movement, emphasis, contrast, symmetry, proportion, space, alignment, pattern, as well as cultural relevance and contextual integration.

However, aesthetics must not always be understood as ornamentation or decoration, but can be grounded in structural integrity and material honesty, as seen in the works of architects like, Frank Lloyd Wright's integration of nature and the use of organic materials in design of Fallingwater, Pennsylvania; or Tadao Ando's Church of the Light in Osaka, Japan, a notable example of honest architecture, where he used concrete as the primary material.

The role of technology - new building materials and construction techniques, also plays an exciting role in developing new aesthetics as a synthesis of technology and art. Italian architect, Pier Luigi Nervi stated, *"the aesthetic sensibility of the designer, who understands their (technology and statistics') intrinsic beauty and validity, welcomes the suggestion and models it, emphasizes it, proportions it in a manner which constitutes the artistic element in architecture"*. Parametricism or the parametric design paradigm leverages software like, Rhino with Grasshopper, Fusion 360, Solidworks, etc. to design built forms of unique visual appeal, as championed by architect Zaha Hadid. Therefore, it is crucial to consider not only functionality, safety, serviceability, and durability but also aesthetics, ensuring proper structural performance throughout the entire lifespan.

5.2.1 Role of Urban Arts Commissions

The Delhi Urban Art Commission (DUAC) was established through a parliamentary act in 1973 with the purpose of advising the Government of India on matters related to preserving, developing, and enhancing the aesthetic aspects of urban and environmental design in Delhi. The commission also provides guidance and recommendations to local bodies regarding construction projects, engineering operations, and development proposals that may impact the skyline, aesthetic quality of the surroundings, or public amenities.

The DUAC plays a crucial role in approving, supervising and recommending projects for development and beautification of urban spaces, conservation and beautification of monuments and green landscaped public areas, preservation and maintenance of heritage monuments and buildings, restoration and redevelopment.

It operates in three capacities: as a policy advisor to the government, a regulatory body, and a think tank. In addition, the following **duties** are assigned to the DUAC as per the Act, 1974;

- a) Development of district centres, civic centres, areas earmarked for Government administrative buildings and for residential complexes, public parks and public gardens.
- b) Re-development of the area within the jurisdiction of New Delhi Municipal Committee including Connaught Place Complex and its environs, Central Vista, the entire bungalow area of Lutyen's New Delhi, and such other areas as the Central Government may, by notification in the Official Gazette, specify.

- c) Plans, architectural expressions, and visual appearance of new buildings in the centres, areas, parks and gardens specified in clauses (a) and (b) including selections of models or statues and fountains therein.
- d) Re-development of areas in the vicinity of Jama Masjid, Red Fort, Qutab, Humayun's Tomb, Old Fort, Tuglakabad and of such other places of historical importance as the Central Government may, by notification in the Official Gazette, specify.
- e) Conservation, preservation and beautification of monumental buildings, public parks and public gardens including location or installation of statues or fountains therein.
- f) Under-passes, over-passes and regulations of street furniture and hoardings.
- g) Location and plans of power houses, water towers, television and other communication towers and other allied structures.
- h) any other project or lay-out which is calculated to beautify Delhi or to add to its cultural vitality or to enhance the quality of the surroundings thereof.
- i) such other matters as may be prescribed by rules.

5.2.2 Heritage Conservation and Structural Repair and Rehabilitation

Structures, be it built recently or of heritage value, deteriorate due to natural weathering or get damaged due to calamities, historic events, etc. and require design and construction interventions. There are various **types of interventions**, such as, *historic preservation, heritage conservation, restoration, rebuilding, rehabilitation, retro-fitting* etc. and each in turn employs different methods to achieve the desired outcome of reviving the original or intended look and feel of the structure.

Heritage Conservation, objectives and guidelines

Architectural conservation, as a discipline is concerned with the “*integrated informed understanding, sustainable care and appropriate renewal and development of the historic environment*”. The aim is to examine the connections between the indoor environment, which is shaped by the interactions among building architecture, materials, structures, services, contents, decorations, and occupants, and the surrounding external environments. It is also the “*process through which the material, historical, and design integrity of any built heritage are prolonged through carefully planned interventions*” (American Conservation Experience). The primary goal is to preserve the original material in its most unchanged state possible, which implies that any repairs or additions made to the built object should not remove, alter, or permanently attach to the original material. It is crucial that all interventions are reversible and removable without causing any damage to the original material, both presently and in the future, and does not involve making artistic choices or experimenting with different materials, i.e., the process of conservation strictly adheres to the object's requirements and specifications, with the sole focus on safeguarding and maintaining the object's authenticity and condition. MoHUA, GoI states that “*Conservation of heritage sites shall include buildings, artifacts, structures, areas and precincts of historic, aesthetic, architectural, cultural or environmentally significant nature (heritage buildings and heritage precincts), natural feature areas of environmental significance or sites of scenic beauty*”.



Fig. 5.9 : UNESCO World Heritage Sites in India (2017)

INTACH (Indian National Trust for Art and Cultural Heritage) further states, “*Conservation of architectural heritage and sites must retain meaning for the society in which it exists. This meaning may change over time, but taking it into consideration ensures that conservation will, at all times, have a contemporary logic underpinning its practice. This necessitates viewing conservation as a multi-disciplinary activity*”.

The **conservation objectives** relevant for India, compiled by INTACH, are as below;

- **Retain Visual Identity:** Preserving the distinct visual identity of a place, created by its architectural heritage and sites, is crucial amidst global homogenization. This preservation should not mimic legally protected monuments but rather adapt to contemporary society while maintaining heritage relevance. Balancing the needs of heritage and societal changes ensures integration into daily life, while controlling visual clutter like advertisements, cables, and antennas enhances the architectural heritage. Additions, such as, street furniture and signage contribute to appreciating the heritage.
- **Adaptive Reuse:** Reusing historic buildings and neighbourhoods is economically viable and conserving architectural heritage by involving traditional craftspeople is effective. Prioritizing continuity of original functions, any new use should consider its impact on the local context, adhering to the capacity and vulnerability of the heritage. Comprehensive documentation should accompany any changes to the original fabric, ensuring coherence and engagement with traditional materials and skills. When altering internal functions, the external Fig. should be retained, involving the local community and communicating the benefits of the changes.
- **Restoration/ Replication/ Rebuilding:** Restoration reinstates the integrity or completeness of architectural heritage/site, aiming to effectively convey its meaning, and may involve reassembly or replacement of missing or deteriorated parts, preceded and followed by documentation. Replication can be appropriate to conserve historic buildings and encourage traditional building methods. Rebuilding at the urban level enhances the visual and experiential quality of the built environment, countering global homogeneity. Reconstruction should be based on minimal physical evidence supported by local crafts people's knowledge and folklore, reinforcing the heritage's bond with society.
Other sources (URA, Singapore) state that the core principle of conservation that applies to all preserved buildings, regardless of their size or complexity, is based on the **concept of maximum retention, sensitive restoration, and careful repair**, often referred to as the "3R"s. Replacement of original elements should only be considered when absolutely necessary, while complete reconstruction contradicts established international conservation practices. When upgrading and adapting a building for new purposes, the existing structure should be preserved by strengthening and repairing its structural elements. Any alterations or reinforcements to these elements should be carried out in a manner that is sympathetic and unobtrusive, utilizing original methods and materials whenever feasible.
- **Employment Generation:** Conservation strategies should focus on employing local craftsmen, labour, and materials to sustain traditional building practices economically. Utilizing architectural heritage as an alternate strategy meets contemporary needs and supports traditional ways of building.
- **Local Material and Traditional Technology:** Preference should be given to local materials and traditional technologies, based on available traditional knowledge systems. Modern substitutes should be considered only if proven efficient and judicious, without compromising the integrity of local building traditions. However, caution must be exercised in using certain

materials that may damage ecological systems, such as shell lime in coastal areas or wood in general, which may require appropriate substitutions.

- **Integrated Conservation:** Architectural heritage conservation should be integrated with social and economic aspirations of the community. Conservation-oriented development should be prioritized, necessitating multi-disciplinary teams that include social workers to facilitate dialogue and decision-making, considering diverse social aspirations.
- **Sustainability:** Conservation should aim to sustain the buildings and/or the traditional skills and knowledge systems for building, positively contributing to the quality of life for the local community over time.

Several Conservation guidelines are available from INTACH, ICOMOS (international council on monuments and sites), and International Centre for the Study of the Preservation and Restoration of Cultural Property (ICCROM), UNESCO.

Repair and Rehabilitation of Structure

Rehabilitation is the process of restoring the structure to service level, once it had and now lost, strengthening consists in endowing the structure with a service level, higher than that initially planned by modifying the structure not necessarily damaged structure. **Repair** is the process of restoring something that is damaged or deteriorated or broken, to good condition, and is a means to perform rehabilitation through the improvement or modification of a structure, partly or wholly, which is damaged in appearance or serviceability. The primary objective of Repair and Rehabilitation is to *restore the structure, maximizing its functional usefulness*. It also serves the purpose of adapting the structure to fulfil new functional and other requirements. Repair and Rehabilitation techniques are employed for various reasons in structures, including but not limited to the following: damage due to accidents, deterioration due to Environment effects (Fire, Natural calamities like earthquake, Flood, Tsunami, cyclones, Soil and structure interaction, such as, settlement of soil or soil failure), and new functional or loading requirements requiring modifications to structure.

The general approach followed in the **Repair and Rehabilitation** process involves;

- **Identifying the building** in need of rehabilitation by performing a Structural Audit of the building.
- **Gathering information** about its history, conducting a preliminary survey that includes necessary tests to evaluate various retrofitting options, materials, feasibility and economy.
- **Identifying the problems** by performing structural calculations and capacity demand ratio for structural members.
- **Conceptualising appropriate and feasible solutions**, such as, retrofitting/construction, that aligns with the building's topographical conditions.
- **Execution of the solution** by getting the rehabilitation of the building done.
- **Post repair/retrofitting tests.**

Retrofitting

Retrofitting refers to the engineering process of modifying existing buildings to improve their structural behaviour while preserving their fundamental intended use. It is important to consider the following aspects for retrofitting, as recommended by (Chandar, 2014);

- **Functionality aspect:** The basic function/ operation of the structure should not be hampered.
- **Structural safety Aspect:** The susceptibility of the structure to an earthquake event has to be within acceptable standards.
- **Importance Level Aspect:** Historic buildings with immense archaeological importance are sometimes beyond the cost factor for retrofitting. Such structures have to be rehabilitated without changing its elegance.
- **Construction Methodology Aspect:** The retrofitting has to be performed using latest construction techniques that have the minimal impact on usual functioning of the buildings.
- **Economy Aspect:** The entire cost of construction has to be practical and logical towards extended life of the structure.
- **Skilled labour availability:** The retrofitting practices need unusual construction method and is highly technical job and calls for utmost care to implement it. A very skilled workmanship must be provided to instrument the suggested measures.

Below are the commonly used **techniques for retrofitting** a building:

1. Installation of additional **Shear Walls** is usually in the exterior of non-ductile reinforced concrete buildings. The wall may be cast-in-place or pre-cast.
2. Addition of **Steel Bracing** is done for higher strength and stiffness when large openings are required, usually for natural lighting.
3. Application of **Wall Thickening Technique** to existing walls of a building is done by adding bricks, concrete and steel reinforcement, and is designed special conditions.
4. Implementation of **Base Isolation Technique** of the superstructure from the foundation is done as a passive structural vibration control technique. It buffers seismic loads and protect the building from damage, however, it is inefficient in high-rises.
5. Adoption of **Mass Reduction Technique**, such as removal of a storey, is done to decrease the loading.
6. Utilization of **Jacketing Method** to strengthen columns and beams by adding a jacket of longitudinal and traverse reinforcement, along with cement, around the structural element.
7. Incorporation of **Fibre Reinforced Polymer (FRP)** to axially strengthen columns and enhance its ductility, and in turn, increase load carrying capacity and improve shear capacity of the member.

8. Application of **Epoxy Injection Method** to non-moving cracks in concrete walls, slabs, columns, and piers, thereby restoring the strength of the concrete.
9. Implementation of **External Plate Bonding** for increasing the shear strength of reinforced concrete beams by completely or partially wrapping steel plates at the joint of a column and beam with external plates or strips, improves the shear strength.
10. **Section-enlarging Reinforcing Method** is used to increase the bearing capacity and cross-section stiffness, widely used for RCC beam, slabs, columns.



Fig. 10: Retrofitting Solutions (Top Left to Right) Shear Wall, Steel Bracing, Section Enlarging reinforcement, (Middle Left to Right) FRP Reinforced, (Bottom Left to Right) Epoxy Injection Method, External plate Bonding

5.3 INNOVATIONS AND METHODOLOGIES FOR SUSTAINABILITY

Various national and international codes and standards are being developed and updated, grounded on methodologies for optimisation, efficiency and further, integrated with sustainability considerations. These entail for specific civil engineering tasks, design requirements and specifications, construction techniques, environmental management and overall, sustainable infrastructure development.

5.3.1 Building Codes

The International Building Code (IBC), developed by the International Code Council (ICC), is widely used and adopted across the United States and the continent of North America. It also serves as the basis for the legislative building codes for several countries, like Mexico and Abu Dhabi. It focuses on adequate light and ventilation, energy conservation, means of egress facilities, and safety to life and property from fire, explosion, and other hazards; etc. India's indigenous building code, the **National Building Code** is developed by BIS.

The **NBC 2016**, has incorporated a lot of new and salient features (Annexure 2) in keeping with the current automation trends and sustainability goals, a few of which are as follows ;

- Detailed provisions relating to requirements for **accessibility in buildings and built environment** for persons with disabilities and the elderly.
- Norms for **solar energy** utilization.
- **Fire and life safety in modern complex buildings** including the high rises, glazed buildings, atria, commercial kitchen and car parking facilities.
- Updated **structural design provisions for wind and seismic loads**, imposed load due to helipad, and blast loads, for safe design and construction of buildings with due focus on ductile detailing.
- Latest research and development inputs and provisions on concrete, steel and masonry buildings with a view to ensuring **disaster resilient buildings**.
- Updated provisions on engineered use of **bamboo in housing** and other building construction.
- Promotion of use of **agricultural and industrial wastes** including **construction and demolition wastes** in building construction without compromising the quality and safety.
- **New and alternative building materials, and technologies** for building construction such as, reinforced masonry, confined masonry building construction and masonry wall construction using rat-trap bond.
- Inclusion of modern lighting techniques such as **LED and induction light** and their energy consumption.
- Provisions on aviation obstacle lights; **electric vehicle charging** and car park management.
- Use of **refrigerants** for **air** conditioning **addressing** **zero** **ozone** **depletion** **potential (ODP)** and ultra-low global warming potential (**GWP**).
- Inclusion of new and **energy efficient options of air conditioning**, heating and mechanical ventilation, such as variable refrigerant flow system, inverter technology, district cooling

system, hybrid central plant using chilled beams, radiant floor components, and geo-thermal cooling and heating.

- Thrust on **envelope optimization** using energy modelling, day lighting simulation, solar shade analysis and wind modelling software to optimize the air conditioning load.
- Air conditioning, heating, and ventilation (HVAC) provisions considering **adaptive thermal comfort conditions** for energy efficiency.
- Updated provisions on **building automation system** to include the latest practices for web-based monitoring and control of performance parameters.
- New chapter on **information and communication enabled installations** in buildings.
- Updated provisions on **rainwater harvesting**.
- New chapter on **solid waste management** covering various solid waste management systems within the building and building complexes.
- Promoting quality of outdoor built environment through updated provisions on **landscape planning**, design and development.
- Promoting **sustainability** in buildings and built environment in tandem with relevant sustainable development goals.
- New chapter on **asset and facility management** to cover provisions relating to management of building assets and associated services, also covering responsibilities of occupants for maintenance of facilities, such as structures, equipment and exterior property.

Another complimentary body is the Indian Society of Heating, Refrigerating and Air-Conditioning Engineers (ISHRAE), which is the only national body in India which has developed **code for IEQ of buildings**, as stated in their 2015 position paper, and has further emphasized on the need for an integrated design approach that blends IEQ and Energy efficiency.

5.3.2 Building Information Modeling Standards

ISO 19650 standard is an internationally recognized guideline for effectively managing information throughout the entire lifecycle of a built asset using building information modeling (BIM). It encompasses the same principles and high-level requirements as the UK BIM Framework and aligns closely with the existing UK 1192 standards.

It is a series of international standards that provide guidelines and requirements for the organization and management of information within the construction industry, including civil engineering projects, by promoting the use of Building Information Modeling (BIM) and standardized processes for information management. It covers various aspects of information management, such as the creation, exchange, and management of digital information throughout the lifecycle of a construction project, with the aim to improve collaboration, efficiency, and interoperability among project stakeholders via BIM. ISO 19650 also addresses topics such as data classification, information security, and documentation requirements. The ISO 19650 standard series includes ;

- **BS EN ISO 19650-1:** This part focuses on the organization and digitalization of information related to buildings and civil engineering works, including building information modelling. It outlines the concepts and principles of information management using BIM.
- **BS EN ISO 19650-2:** This part deals with the organization and digitalization of information during the delivery phase of assets in buildings and civil engineering works. It provides guidance on information management using BIM during this stage.
- **BS EN ISO 19650-3:2020:** This part addresses the organization and digitalization of information in the operational phase of assets in buildings and civil engineering works. It covers information management using BIM during this period.
- **BS EN ISO 19650-5:2020:** This part focuses on the organization and digitalization of information, including building information modelling, with a security-minded approach to information management.

These standards are founded on the UK's standards for information management using building information modelling, namely BS 1192:2007 + A2:2016 and PAS 1192-2:2013. PD 19650-0 - UK Transition Guidance, which will along with the UK National Forewords and National Annex aid implementation of the ISO standards in the UK

5.3.3 Environment Management System Standards



An **Environmental Management System** is “a framework that helps an organization achieve its environmental goals through consistent review, evaluation, and improvement of its environmental performance” (EPA). It enables organizations to address their regulatory obligations in a systematic and cost-effective manner, proactively reduces environmental risks and enhances health and safety practices for both, occupants, and the public. Additionally, an EMS can tackle non-regulated concerns like, energy conservation, while promoting better operational control and encouraging stakeholder stewardship.

The **fundamental components of an Environmental Management System** encompass;

1. Assessing the organization's environmental objectives.
2. Analysing environmental impacts and compliance obligations, including legal and other requirements.
3. Establishing environmental objectives and targets aimed at reducing environmental impacts and meeting compliance obligations.
4. Implementing programs to achieve these objectives and targets.
5. Monitoring and measuring progress towards the established objectives.
6. Ensuring employees are well-informed and possess the necessary environmental awareness and competence.

7. Conducting regular reviews to track progress and identify opportunities for improvement.

The **ISO 14001:2015** is a standard that establishes the requirements for an Environment Management System to enable organizations to improve their environmental performance, fulfilment of compliance obligations and achievement of environmental objectives. ISO 14001:2015 is applicable to organizations of all sizes, types, and industries, and covers the environmental aspects of an organization's activities, products, and services that the organization can control or influence taking a life cycle perspective into account. It can be utilised either in its entirety or partially, to systematically enhance their environmental management practices, however, it does not establish specific environmental performance criteria.

5.3.4 ISO TC268 'Sustainable Cities and Communities'

TC 268 in the area of **Sustainable Cities and Communities** focuses on standardization efforts that involve developing requirements, frameworks, guidance, and tools to promote sustainable development, with considerations for smartness and resilience. The aim is to support all cities, communities, and stakeholders in rural and urban areas in their journey towards increased sustainability, in line with the SDGs. It has several working groups in the areas of infrastructure metrics, Integration and interaction framework for smart community infrastructures, Data exchange and sharing for smart community infrastructures, Power plant, Disaster risk reduction and Utility tunnel.

ISO/TC268, is responsible for the ISO 37100: 2016 series of standards that assists cities in defining their sustainability objectives and implementing strategies to achieve them. ISO/TR 37150 introduced indicators like Global City Indicators, Green City Index, and Smart City concepts driven by ICT. Efforts are underway to enhance existing indicators, such as, ISO/NP 37122, which focuses on "Sustainable Development in Communities - Indicators for Smart Cities" through legislations (currently in the proposal phase, and soon to be replaced by the ISO/AWI 37100)

The world is constantly developing new knowledge, methodologies, and innovative tools to strategically address improvements in the built environment and strive towards sustainability, due to its profound impact – at the societal and global level. It is imperative that the civil engineer of the future is much more than a master builder, but must gain competence in several other areas of leadership to ensure that the aim of sustainable development is achieved.

UNIT SUMMARY

This elementary unit presents the three pronged story of a good built-environment, as described by Vitruvius – utility, durability and beauty (aesthetics), from the lens of sustainable development. The unit elaborates the various sustainability practices and applications within the domain, such as, Facility management and sustainability strategies, Building control systems, Building certification and rating, and Smartness readiness Index, to enable the civil engineer for sound environmental decision-making. Additionally, the unit further introduces the importance and relevance of ‘aesthetics’ and its need to preserve the culture inherent in built-environment through practices of heritage conservation, and repair and rehabilitation. Lastly, the knowledge of the various national and international codes and standards for ensuring prioritisation and incorporation of environmental sustainability in practice is propounded.

EXERCISES

I. Multiple Choice Questions

- Q. 5.1 Which of the following are not key activities of Facility Management?
- (a) Maintenance, cleaning, testing, and inspection
 - (b) Regulatory Compliance
 - (c) Heritage Conservation
 - (d) Safety and Security
- Q. 5.2 What are practical strategies of sustainable facility management?
- (a) Waste management
 - (b) Green Building Certification
 - (c) Performance Monitoring
 - (d) all of the above
- Q. 5.3 Which is not a design strategy for reduction of embodied energy?
- (a) Use of recyclable materials
 - (b) Parametricism
 - (c) Design for deconstruction
 - (d) Low maintenance design

Q. 5.4 Which LEED category has the highest associated credit points (33 credits)?

- (a) Location and Transportation
- (b) Materials and Resources
- (c) Energy and Atmosphere
- (d) Water efficiency

Q. 5.5 Name India's national body responsible for developing contextually appropriate Heritage Conservation guidelines and objectives?

- (a) INTACH
- (b) ICOMOS
- (c) ICCROM
- (d) UNESCO

Answers of Multiple Choice Questions: 5.1 (c) , 5.2 (d) , 5.3 (b), 5.4 (c), 5.5 (a)

II. Short and Long Answer Type Questions

Q. 5.6 What is Aesthetics? Why is it an important consideration in built-environment.

Q. 5.7 What is a Building Control System? Illustrate the different types of sub-systems that it may have and their functionalities.

Q. 5.8 Discuss the various Green Building Certification/Rating systems used in India. Discuss one in detail.

Q. 5.9 What is embodied energy and embodied carbon? What are some of the design strategies for reduction of embodied energy and embodied carbon, elucidate with examples.

Q. 5.10 What are some of the new and innovative additions to the NBC 2016? Explain any two that will have profound impact on environmental sustainability.

KNOW MORE

History of Facility Management



<https://www.rentokil-pestcontrolindia.com/facilities-management/history/>

About Embodied Carbon



<https://worldgbc.org/advancing-net-zero/embodied-carbon/>

Environmental Management System



<https://www.epa.gov/ems/learn-about-environmental-management-systems#costs>

About Smart Growth: What is Built Environment?



<https://www.epa.gov/smartgrowth/about-smart-growth#what-is-the-built-environment>

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Unit 5 – Built Environment

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